

ICPSR 4283

**National Center for Early
Development and Learning
Multistate Study of
Pre-Kindergarten, 2001-2003**

User Guide

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National Center for Early Development and Learning Multistate Study of Pre-Kindergarten, 2001-2003

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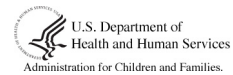
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National Center for Early Development and Learning

Multi-State Study of Pre-Kindergarten

Study Description & Documentation for Public Release of Data

www.ncedl.org

July 1, 2005

**National Center for Early Development and Learning
Multi-State Study of Pre-Kindergarten
Study Description**

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National Significance

In recent years, schools have become increasingly involved in providing services for children and families prior to entry into formal school at kindergarten. In fact, over the last decade, the public investment in formal, school-related programs for the education of young children has soared. This increase is expected to continue for the foreseeable future, yet little is known about the nature and quality of the programs offered by or through schools, much less the impact of these growing programs on the children and families being served. The National Center for Early Development and Learning (NCEDL) sought to address these issues and, by doing so, to provide guidance to state and local policy-makers as well as to the local personnel implementing the pre-kindergarten programs.

Of particular interest to NCEDL were the numerous state early childhood education initiatives operated under the auspices of the state education agency or in conjunction with the state agency. Efforts to document these state pre-kindergarten programs have been made by the National Institute for Early Education Research (NIEER; Barnett, Hustedt, Robin, & Schulman, 2004), Children's Defense Fund (Blank, Schulman, & Ewen, 1999), the National Center for Children in Poverty (Knitzer & Page, 1998), the Families and Work Institute (Mitchell, Ripple, & Chanana, 1998), and Yale's Bush Center (Ripple, Gilliam, Chanana, & Zigler, 1999). However, no previous study included classroom observations and/or direct assessments of children's skills across a large sample of programs in multiple states.

Key challenges for national and state policy-makers and local practitioners include decisions about 1) whether to provide pre-kindergarten programs, 2) to whom such programs should be available, 3) how the services should be financed, 4) what the goals of these programs are, and 5) what models and practices should be used. Currently little information is available about any of these aspects of this fast-growing

and emerging sector of educational services. Professionals disagree on such critical issues as the extent to which academic skills should be taught, the amount and type of teacher training required of staff, the role of families in the programs, and the intensity needed to achieve desired results. Thus, it is imperative to examine the nature of the program structure and practice to help guide the expansion of pre-kindergarten services to maximize the benefits for both children and families.

In order to fill the gaps in current knowledge about pre-kindergarten programs, NCEDL conducted a large-scale study of pre-kindergarten in six states. For this study, “pre-k” included center-based programs for 4-year-olds that were fully or partially funded by state education agencies as well as those that are operated in schools or under the direction of state and local education agencies. The study is generated completely new data about the nation’s pre-k programs.

Primary Research Questions

The NCEDL Multi-State Study of Pre-Kindergarten had two primary purposes:

(1) to describe the variations of experiences for children in pre-kindergarten and kindergarten programs in school-related settings (public schools and state funded pre-k classrooms in community based settings), and

(2) to examine the relations between variations in pre-kindergarten/ kindergarten experiences and children's outcomes in early elementary school.

We are examined both the quality of children's experiences and the nature of teaching and learning practices within these programs. We derive our construct of quality from the child care research tradition. Within this tradition quality is defined both as structure (e.g., class size) and process (e.g., teacher involvement and sensitivity or attention to home language and culture). In this construct, relatively little attention has been paid to the specifics of literacy and numeracy instruction, with primary attention to global issues of relationships and environmental provisions

for learning. We derive our construct of practices from research on teaching and learning in elementary school classrooms. Examining practices includes, for example, a focus on how math and reading are taught or whether relationship-based classroom management practices are used. Special emphasis is placed on literacy, numeracy, and social-emotional domains. We are applying the best from each of these traditions to both pre-kindergarten and kindergarten settings.

The Multi-State Study of Pre-Kindergarten addressed six primary research questions:

(1) What is the nature and distribution of education and experience of teachers and teacher assistants in public school pre-kindergarten programs?

(2) What is the nature and distribution of global quality and specific practices in key areas such as literacy, math, and teacher-child relationships in a diverse sample of public school pre-kindergarten programs for 4-year-olds as well as in a similarly diverse sample of kindergarten classes?

(3) How do quality and practices vary as a function of child and teacher characteristics (e.g., child gender, race, home language, family income, teachers' years of education) and classroom, program, community, and state structural variables (e.g., teacher-child ratio, funding base of the program, teacher salary, degree of state regulation) for children with different demographic characteristics (e.g., race, gender, home language, and family income)?

(4) Do quality and practice vary in relation to combinations of these variables? For example, are quality and practice a function of family poverty and teacher pay or education?

(5) Can children's outcomes at the end of their pre-kindergarten year be predicted by the children's experiences in pre-kindergarten programs? Are the various dimensions of quality and/or practice differentially related to different outcomes? Are

these relationships constant across children with different characteristics (e.g., race, gender, home language, and family income)?

(6) Do pre-kindergarten quality and practices predict children's transition to kindergarten and children's skills at the end of the kindergarten year? Is this transition and/or outcome moderated by child characteristics, like race, gender, and family income?

Additionally, questionnaires sent to the study-children's first grade teachers allow for a limited number of similar questions to be addressed through the first grade. For instance, teacher reports of first grade academic skills allow for consideration of the longer-term effects of pre-kindergarten and kindergarten quality on children's academic skills (as reported by teachers).

State Selection

Forty-one states, plus the District of Columbia, invested some state funds in pre-kindergarten initiatives in 1999. As a first step on selecting states to participate in NCEDL's study, we used existing information (e.g., Seeds of Success, NCEDL's Survey of State Specialists) to learn which states had significant pre-kindergarten initiatives. We limited our pool of potential states to those that served 15% of their state's 4-year-olds or served at least 15,000 4-year-olds. This limited the pool of potential states to roughly 19.

From those 19 states, we selected 13 to participate in a planning meeting in the fall of 2000. These 13 were chosen to represent diversity with regard to: 1) length of program day, 2) teacher credentialing requirements, 3) program locales (in schools versus in community settings), and 4) geography. Of the 13 states invited to attend this meeting, 10 states elected to attend.

This meeting was intended to inform the states about the upcoming study, gather information about their pre-kindergarten initiatives, gauge the interest in

participating in the study, and learn which states had infrastructure in place that would facilitate our data collection efforts (e.g., lists of participating programs that included key sampling variables, ability to aid in following children into kindergarten). Following this meeting, six states were invited to participate in the study. Of the six originally invited states, five agreed. The sixth was replaced so that the study has six participating states: California, Illinois, New York, Ohio, Kentucky, and Georgia. Budget and time constraints prohibited us from randomly selecting from the entire states of California and New York. Instead, we limited our selection to 20 sites in the greater Los Angeles area, 20 sites in California's Central Valley, 20 sites in New York City, and 20 sites within a 50-mile radius of Albany, New York. Thus, in California and New York, the study represents only pre-kindergarten programs in those regions. In the other four states, we randomly selected from programs in the entire state.

In 2001-2002 (when the study began), 43% of all children in the United States who were participating in state-funded pre-kindergarten were in one of these 6 states, and 42% of state dollars spent on pre-k were in one of these six states. **Nonetheless, it is important to note that this sample only reflects the four states and two regions included in the study. It is not meant to be nationally representative.**

Sampling

After state selection, four levels of random sampling took place: zip code, site, classroom, and child. We randomly selected: 20 zip codes from each state/region, 2 sites from each selected zip code, 1 pre-k classroom from each selected site, and 4 pre-k children from each selected classroom. Sample size decisions were based primarily on budgetary factors. Appendix A illustrates the sampling process, the response rates, and the total fall sample sizes.

Zip Code Selection

Zip code was primary sampling unit. This was done to ensure that each participating site had another participating site near-by, thus decreasing the amount of travel required by the data collectors. In more rural states/areas, the first 4 digits of the zip code were used as the PSU. In more densely populated areas, the entire 5-digit zip code was used. Twenty 4- or 5- digit zip codes were selected in each state/region.

Site and Classroom Selection

Sites (i.e., schools or centers) were selected from lists provided by the state education agencies of all centers/schools receiving state pre-k funds (either directly or as subcontractors). Within each site, data collectors worked with the program director to randomly select one classroom to participate. Eligible classrooms had to receive state funds and enroll at least five eligible children (child eligibility requirements are described in the next section).

The goal in drawing the samples of sites and classrooms was to maximize within state, classroom-level diversity with regard to three key variables [1) **programs in school buildings versus not**, 2) **full day versus part-day** programs, and 3) **lead teachers with and without a Bachelor's degree**. States vary widely with regard to the regulations regarding these key variables, leading to markedly different distributions in each state. NCEDL wanted to maximize within state variance on these variables in order to limit potential confounds between state and these three key variables. This goal, each state's distribution on these key variables, and available sampling information led us to different site and classroom sampling strategies in each state.

In general, the site and classroom selection strategies in each state involved oversampling on whichever two key variables (one at the site level and one at the

classroom level) had the most uneven distributions (i.e., furthest from 50%-50% split). Of course, oversampling was not possible in states that had no variation on a particular variable (e.g., in one state, all pre-k classrooms were full-day).

In total, 238 pre-k sites participated in the fall of the pre-kindergarten year and two additional sites joined the study in the spring. To obtain this sample of 240 sites, 335 sites were contacted. Selected sites that were found to be ineligible or declined to participate were replaced by another randomly selected site from the same PSU. Of the 95 that were contacted initially but did not participate, 26 were ineligible (e.g., did not receive state funds, did not serve 4 year-olds), 58 declined, and 11 never responded. Thus, of those sites that were eligible, 78% agreed to participate.

Within each site, one classroom was randomly selected for participation. Of all randomly selected teachers, 16 teachers declined to participate (from 12 sites). Thus, of the teachers selected in the initial random draw, 94% agreed to participate in the study. In cases where the teacher declined, another classroom (and its lead teacher) from the same site was selected at random for participation.

We had limited information on which to compare sites that agreed to participate with those that declined. During sampling, five of the six state agencies provided information about program length (full-day or half-day) and location (in school or not in school) for all sites in their state. In those states, we can use that information to compare sites that agreed to participate with those that declined. The sixth state was not able to provide any site-level information, so no comparisons can be made. In the five states where we have the needed information, the sample sites and the sites that declined were very similar in the length and location. In both the sample and the sites that declined participation, about 50% were half-day programs. In these five states, approximately 59% of the sample (unweighted) compared to 60% of the sites that declined were located in schools.

Child Selection

In total, 940 children participated in the fall of pre-k data collection. Data collectors worked with selected teachers to recruit children into the study. Teachers sent packets home with all children in the classroom. The packets contained 1) a consent form describing the study, 2) a family contact sheet that provided information needed to contact the families, and 3) a short demographic questionnaire. Parents returned these packets to the teacher. On the first day of data collection, the data collectors determined which of the children whose parents had consented were eligible for participation. Eligible children are those who 1) were old enough for kindergarten the following fall (2002), 2) did not have an Individualized Education Plan, and 3) spoke English or Spanish well enough to understand simple instructions according to the teacher. From that group, data collectors randomly selected four children to participate. Whenever possible, two girls and two boys were selected in each classroom. On average, the parents of 61% of eligible children in each room consented to have their child participate.

The Sample Maintenance

Spring of Pre-Kindergarten

All children who had participated in the study during the fall of pre-kindergarten and were still enrolled at the same pre-kindergarten site participated in the spring of pre-kindergarten data collection. Because the focus of this study was on pre-kindergarten, we did not attempt to follow children who left the pre-kindergarten program during the year. In the spring of the pre-k year (2002), study children who had disenrolled from their program ($n = 56$) were replaced with another randomly selected eligible child when possible. If possible, a child of the same gender as the one

who had disenrolled was selected. Also, additional study children were recruited in the spring in classes where fewer than four children participated in fall. In total, 76 children joined the study in the spring. This resulted in 960 children participating in the spring. The variable CHPARTP indicates if the child participated in fall, spring, or both.

A few ($n= 5$) children who participated in the fall of pre-k had moved to a new classroom at the same site during the spring of pre-k. We replaced those children in their original classroom with a newly selected child, and we sought to conduct a spring assessment and collect teacher questionnaires with those children in their new room. We did not attempt to obtain observational data (ECERS-R, CLASS or Snapshot) from those children's new classrooms. These children have a value of '2' on the variable T_CHNGCP.

Kindergarten

The goal for kindergarten data collection was to locate all 960 children who had participated in the spring of pre-k and collect data from them, their teachers, and their parents during the 2002-03 school year. The study-children, who had been in 240 sites during pre-kindergarten, spread to over 800 different classrooms during the kindergarten year. Many children moved during the course of the kindergarten year, resulting in a need to collect data from multiple classrooms for a single child. The variable NUMCLSRMK indicates, for each child, the number of different classrooms from which observational, assessment, or questionnaire data were collected during the kindergarten year.

In the end, complete data¹ were collected from 778 (81%) children and partial data were collected from 132 (14%). No data were collected from 49 (5%) of the children. The variable CHPARTK identifies which children have complete, partial data, and no data (note: this variable is missing for cases where the child did not participate in spring of pre-k data collection because we did not attempt to follow those children into kindergarten). For children with partial or no data, the variable CHNRC DK indicates the reason for the missing data.

First-Grade

During 2003-04 school year (1st grade for most study children), we attempted to collect teacher questionnaires from all children who had participated in the spring of pre-kindergarten data collection (aside from the 33 children whose parent had specifically withdrawn them from the study). The original parent consent had covered only the pre-kindergarten and kindergarten years, therefore new parent consent had to be obtained for each child. We successfully obtained parental consent and collected questionnaires from the teachers of 593 (62%) of the children who had participated in the spring of pre-k data collection.

Generalizability of Weighted Samples.

Technically, our sample can provide representative samples of the Pre-Kindergarten classrooms and children attending those classrooms in Illinois, Ohio, Kentucky, and Georgia and in two regions of California and New York. It is not meant to be nationally representative. The goal was to describe each state, and so the 6 states are equally weighted in the data. All analyses should include state when weights are used because the samples were drawn separately for each state and the weights add to 100% within state, not across state.

¹ Complete data is defined as at least: 1 ECERS, 1 Snapshot, 1 CLASS, fall and spring child assessments, fall and spring teacher questionnaires, and fall and spring teacher questionnaires about the study child. Most children also

Post-stratification Weights

In order to adjust the weights to represent children lost in the Spring of Pre-K follow-up, post stratification re-weighting was performed. For children within a classroom, because boys and girls were sampled, and the dispersions were sometimes low, they were considered sampled without replacement, by gender. In classrooms where children were replaced, the weights for all the children in the classroom are changed to reflect this new dispersion.

There were 49 children lost from Spring of Pre-K to Kindergarten. Only children were followed, therefore, there is no need to adjust for non-response of the classroom. The child weights were adjusted for non-response using a model based approach. Children were divided based upon important demographic characteristics and then proportionally re-weighted within each cell that contained attrition.

The weights should be used any time that cross-sectional analysis within a period is performed. For example, the effect on children's test scores in the Spring of Pre-K based upon classroom and teacher characteristics should use the Spring of Pre-K child weights. So on and so forth.

Measures

Appendices B, C, & D contains charts showing the measures that were collected at each time point. See the actual measures that accompany this description for complete information and exact items.

Data Collection Schedule

Appendices A, B, & C indicate the intended frequency of administration of each measure. Snapshot and CLASS observations were always conducted on the same days, by the same observer. Observers took breaks during the Snapshot observations

have a parent questionnaire and/or an end of year questionnaire, but these are not required to be considered "complete."

at regularly scheduled intervals to code the CLASS. During pre-k, observation days (ECERS-R or Snapshot/CLASS) lasted from the beginning of class until the end of class in part-day rooms and from the beginning of class until children were settled for nap in full-day rooms. Observers stayed with the children all day, even if they left the classroom for lunch, outside time, or specials.

In kindergarten, the observation schedule was slightly different, because the classroom days were generally longer and observers needed breaks. Snapshot/CLASS observations lasted the entire class day (from beginning to end in both full- and part-day rooms), but no observations were made during lunch, any recess immediately following lunch, and nap (if any) and those times are not counted in the denominators for the various time percentages. **For this reason, pre-k and kindergarten Snapshots should not be compared (especially for meals/snacks, gross motor, recess/outside time).**

Kindergarten ECERS-R observations all lasted the entire day. Observers took a 30 minute lunch break at a time that they felt would not interfere with the accurate rating of the items.

During the pre-k year, the ECERS-R was generally conducted by a different individual from the Snapshot/CLASS, on one of the same days (Snapshot/CLASS lasted two days, ECERS-R lasted one). During the kindergarten year, the ECERS-R was generally done by the same person as the Snapshot/CLASS, on a different day.

Data Collector Training and Reliability

Most data collectors (20 out of 24) remained with the project over the 2 years of observational data collection (pre-k and kindergarten). Data collectors from all states were trained on all measures in a group training during the summer prior to pre-k data collection and again during the summer prior to kindergarten data collection. After training they practiced the observations with their data collection partner in

several classrooms and practiced the child assessment with volunteer children. Before each data collection period (fall of pre-k, spring of pre-k, fall of Kindergarten, spring of kindergarten) they met reliability criteria on the observational measures. Appendix E shows the type of reliability testing that was conducted at each time point and the level of reliability attained.

Between formal tests, data collector participated in regular phone calls with supervisor and informal reliability tests (generally a couple videotaped segments followed by discussions with goldstandards). When reliability problems were observed additional follow-up was conducted.

Data collectors videotaped themselves completing an assessment with a 4-year-old child prior to pre-k data collection. Each videotape was reviewed by a project supervisor who determined whether the data collector was following the assessment protocol. If the data collector did not follow the protocol, errors were noted on a 90-item checklist developed for the assessor certification, feedback was given to the data collector, practice prescribed, and another tape submitted until the standards were met. Prior to fall of kindergarten data collection, a supervisor watched each data collector conduct the assessment with a young child and again either “passed” the data collector, or provided feedback, prescribed practice and re-tested later. Data collectors completed written tests about assessment protocols prior to spring data collection each year.

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Peer-reviewed journal articles (published and in press)

Barbarin, O., Bryant, D., McCandies, T., Burchinal, M., Early, D., Clifford, R., Pianta, R., & Howes, C. (in press). Children enrolled in public pre-k: The relation of family life, neighborhood quality, and socio-economic resources to early competence. *American Journal of Orthopsychiatry*.

Clifford, R. M., Barbarin, O., Chang, F., Early, D. M., Bryant, D., Howes, C., Burchinal, M., & Pianta, R. (in press). What is pre-kindergarten? Characteristics of public pre-kindergarten programs. *Applied Developmental Science*.

La Paro, K. M., Pianta, R. C., & Stuhlman, M. (2004). The Classroom Assessment Scoring System: Findings from the pre-k year. *The Elementary School Journal*, 104(5), 409-426.

Pianta, R., Howes, C., Burchinal, M., Bryant, D., Clifford, R. M., Early, D. M., & Barbarin, O. (in press). Features of pre-kindergarten programs, classrooms, and teachers: Prediction of observed classroom quality and teacher-child interactions. *Applied Developmental Science*.

Manuscripts under review

Barbarin, O. A, McCandies, T., Early, D., Clifford, R. M., Bryant, D., Burchinal, M., Howes, C., & Pianta, R. (under review). School readiness: The perspectives of families of children enrolled in public sponsored Pre-K programs. *Early Childhood Research Quarterly* (submitted January, 2005).

Barbarin, O. A, McCandies, T., Early, D., Clifford, R. M., Bryant, D., Burchinal, M., Howes, C., & Pianta, R. (under review) Quality of Pre-K: What Families are looking for in Public sponsored programs. *Early Education and Development*. (submitted February, 2005).

Clifford, R. M., Early, D. M., Bryant, D. M., Barbarin, O., Burchinal, M., Pianta, R. C., Ritchie, S., & Howes, C. (under review). Early findings from the NCEDL Pre-Kindergarten Study. *Principal*. (submitted May, 2005).

Early, D. M., Bryant, D., Pianta, R., Clifford, R., Burchinal, M., Ritchie, S., Howes, C., & Barbarin, O. (under review) Are teacher education, credentials, and workshop participation related to classroom quality and children's academic gains in pre-kindergarten? *Early Childhood Research Quarterly* (submitted December, 2004).

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Howes, C., Burchinal, M., Pianta, R. Bryant, D., Early, D. M., Clifford, R., Barbarin, O & (under review). Ready to learn? Children's pre-academic achievement in pre-kindergarten programs. *Developmental Psychology* (re-submitted in May, 2005)

Other publications

Bryant, D., Clifford, R. M., Saluja, G., Pianta, R., Early, D., Barbarin, O., Howes, C., & Burchinal, M. (2002). *Diversity and directions in state pre-kindergarten programs*. Chapel Hill: The University of North Carolina, FPG Child Development Institute, NCEDL. Available:
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A. Sampling Plan

**19
preliminary
states**

States selected to meet threshold for numbers or percentage of children served.

6 states

States selected to be diverse with regard to:

- Teacher credential requirements
- Location of classrooms (in vs. out of school)
- Intensity (length of day/year)
- Geography

**40 sites in each
state**

Local schools/centers selected through stratified random sampling to maximize within state variation in:

- Teacher degree (Bachelor's vs. no Bachelor's)
- Location of program (in vs. out of school)
- Intensity (Full-day vs. part-day)

78% of eligible sites agreed to participate

Fall of pre-k $n = 238$

**1 classroom per
site**

Classrooms selected using stratified random sampling to maximize within state variation in:

- Teacher credentials
- Intensity (length of day)

94% agreed to participate

Fall of pre-k $n = 238$

**4 children per
classroom**

Children selected randomly; half girls, half boys

61% of parents of eligible children consented

Fall of pre-k $n = 940$

B. Pre-Kindergarten Instruments: Fall 2001-Spring 2002

		Instrument	Author(s)
Parent Questionnaire	Fall 2001	Child and family demographics (English and Spanish versions)	NCEDL
Teacher Questionnaire	Fall 2001	Teacher demographic characteristics (Items 1-14)	NCEDL
		Class/student characteristics (Items 15-27)	NCEDL
		Curriculum Items (Items 28-30)	NCEDL
		Ideas about Children (Modernity Scale) (Item 31)	Schaefer & Edgerton
		Transition Activities (Item 32)	Pianta
		Teacher Self-Efficacy (Item 33)	Bandura (revised by NCEDL)
		Your Feelings (Depression Index) (Item 34)	Radloff
Teacher Questionnaire	Spring 2002	Class/student Characteristics (Items 1-12)	NCEDL
		Assessment (Items 13-14)	NCEDL
		Teacher training (Item 15)	NCEDL
		Working with Parents (Items 16-19)	NCEDL
		Assistant/co-teacher (Items 20-22)	NCEDL
		Transition Activities (Item 23)	Pianta
		Your Feelings (Item 24)	Radloff
Teacher Questionnaire about Study Child	Fall 2001	Child's Health (Item 5)	
		Teacher Child Rating Scale (Social Skills) (Items 6 & 7)	Hightower
		Severity of Mental Health Issues (Items 8-13)	Barbarin
		Language and Literacy Skills (Academic Rating Scale) (Item 15)	ECLS-K

		Instrument	Author(s)
Teacher Questionnaire About Study Child	Spring 2002	Child's Health (Item 3)	
		Teacher Child Rating Scale (Social Skills) (Items 4 & 5)	Hightower
		Severity of Mental Health Issues (Items 6-12)	Barbarin
		Special Services (IEP, disability) (Items 13-18)	NCEDL
		Language and Literacy Skills (Academic Rating Scale) (Item 20)	ECLS-K
		Mathematical Thinking (Academic Rating Scale) (Item 21)	ECLS-K
		Student-Teacher Relationship (Item 22)	Pianta
		Home-School Relationships (Items 23-29)	Barbarin
		Family Involvement (Item 30)	Barbarin
End of Year Teacher Questionnaire	Late Spring 2002	IEP (Items 4-5)	NCEDL
		Absences (Item 6)	NCEDL
		Class size, length of school year (Items 7 & 8)	NCEDL
		Transition Activities (Item 9)	Pianta
Administrator Questionnaire	Spring 2002	School/program characteristics (Item 1-11)	NCEDL
		Services Provided (Item 12)	NCEDL
		Pre-K program goals (Item 13)	NCEDL
		Curriculum (Items 14-18)	NCEDL
		Transition Activities (Items 19-20)	Pianta
		Pre-K Teacher Characteristics (Items 21-22)	Pianta
		Ideas about Children (Modernity Scale) (Item 23)	Schaefer & Edgerton
		Funding & Teacher Wages and Benefits (Items 24-26)	NCEDL
		Program Eligibility & Recruitment (Items 27-30)	NCEDL
		Parent Involvement (Items 31—36)	Barbarin
		Home Visiting (Items 37-38)	NCEDL
		Administrator Demographic Characteristics (Items 39-51)	NCEDL

		Instrument	Author(s)
Classroom Observation	1 day, Fall 2001 & 1 day Spring 2002	Early Childhood Environment Rating Scale- Revised	Harms, Clifford, & Cryer
	2 days, Fall 2001 & 2 days Spring 2002	Emerging Academic Snapshot	Ritchie, Howes, Kraft-Sayre, & Weiser, 2001
		Classroom Assessment Scoring System	Pianta, La Paro, & Hamre, 2004
Child Assessment	Fall 2001 & Spring 2002	Pre-LAS (given only to children who do not speak English at home)	Duncan & De Avila
		PPVT-III (English only)	Dunn & Dunn
		TVIP (Spanish only)	Dunn, Padilla, Lugo & Dunn
		Naming Letters (English and Spanish versions)	Bryant, Barbarin & Aytch
		Naming Numbers (English and Spanish versions)	Bryant, Barbarin & Aytch
		Counting (English and Spanish versions)	Bryant, Barbarin & Aytch
		Color Names (English and Spanish versions)	FACES
		OWLS Oral Expression Scale (English only)	Carrow-Woolfok
		CTOPP -- Elisions Subtest (English only)	Wagner, Torgesen, & Rashotte
		Woodcock Johnson – Sound Awareness (English only)	Woodcock, McGrew & Mather
		Woodcock Johnson – Applied Problems (English only)	Woodcock, McGrew & Mather
		Woodcock Munoz – Problemas Aplicados (Spanish only)	Woodcock & Munoz-Sandoval
		NCEDL Early Writing Task (English and Spanish versions)	Bryant
		Test Session Observations	Woodcock, McGrew & Mather

C. Kindergarten (2002-2003) Instruments

		Instrument	Author
Teacher Questionnaire	Fall of K 2002	Teacher Demographic Characteristics (Items 2-13)	
		Student/class Characteristics (Items 14-29)	
		Curriculum (Items 30-36)	NCEDL
		Student., Assistant, Co-Teacher (Items 37-39)	
		Ideas about Children (Modernity Scale) (Item 40)	Schaefer & Edgerton
		Transition Activities (Item 41)	Pianta
		Teacher Self-Efficacy (Item 42)	Bandura
		Coping with Children's Negative Emotions Scale (Items 43-46)	Fabes, Eisenberg, & Bernzweig
		Your Feelings (Depression Index) (Item 48)	Radloff
	Spring of K 2003	Class/student characteristics (Items 1-13)	NCEDL
		Assessment (items 14-15)	NCEDL
		Working with Parents (Items 17-20)	NCEDL
		Assistant/co-teacher (Items 21-24)	NCEDL
Teacher Questionnaire about Study Child	Fall of K 2002	Child Characteristics & Health (Items 4-7)	
			Hightower
		Severity of Mental Health Issues (Items 10-15)	Barbarin
		Special Services (IEP, disability) (Items 16-20)	NCEDL
		Language and Literacy Skills (Academic Rating Scale) (Item 22)	ECLS-K
		Mathematical Thinking (Academic Rating Scale) (Item 23)	ECLS-K
	Spring of K 2003	Child's Health (Item 3)	
		Teacher-Child Rating Scale (Social Skills) (Items 4 & 5)	NCEDL
		Severity of Mental Health Issues (Items 6-12)	Barbarin
		Special Services (IEP, disability) (Items 13-18)	NCEDL
		Language and Literacy Skills (Academic Rating Scale) (Item 21)	ECLS-K
		Mathematical Thinking (Academic Rating Scale) (Item 22)	ECLS-K
		Student-Teacher Relationship (Item 23)	Pianta
		Home-School Relationships (Items 24-30)	Barbarin
		Family Involvement (Item 31)	Barbarin

		Instrument	Author
Classroom Observation	1 time, Winter of 2002-03	Early Childhood Environments Rating Scale – Revised (ECERS-R)	Harms, Clifford, & Cryer
	3 times across school year, at least 4 weeks apart	Emergent Academic Snapshot	Ritchie, Howes, Kraft-Sayre, & Weiser, 2001
		Classroom Assessment Scoring System (CLASS)	Pianta, La Paro, & Hamre, 2004
Child Assessment	Fall 2002 & Spring 2003	Pre-LAS (given only to children who did not pass the Pre-LAS during the previous assessment)	Duncan & De Avila
		Peabody Picture Vocabulary Test-III (PPVT-III) (given to children assessed in English and Spanish)	Dunn & Dunn
		Test de Vocabulario de Imágenes Peabody (TVIP) (Spanish only)	Dunn, Padilla, Lugo & Dunn
		Naming Letters (English and Spanish versions)	Bryant, Barbarin & Aytch
		Naming Numbers (English and Spanish versions)	Bryant, Barbarin & Aytch
		Counting (English and Spanish versions)	Bryant, Barbarin & Aytch
		Color Names (Spanish only)	FACES
		OWLS Oral Expression Scale (English only)	Carrow-Woolfok
		CTOPP -- Elisions Subtest (English only)	Wagner, Torgesen, & Rashotte
		Woodcock Johnson – Sound Awareness (English only)	Woodcock, McGrew & Mather
		Woodcock Johnson – Letter Word Identification (English only)	Woodcock, McGrew & Mather
		Woodcock Johnson – Applied Problems (English only)	Woodcock, McGrew & Mather
		Woodcock Munoz – Problemas Aplicados (Spanish only)	Woodcock & Munoz-Sandoval
		NCEDL Early Writing Task (English and Spanish versions)	Bryant
		Test Session Observations	Woodcock, McGrew & Mather

		Instrument	Author
Parent Questionnaire	Spring of K 2003	Transition to Kindergarten (items 3a-e)	Pianta
		Relationship with Child's School (items 4-8)	Barbarin
		Home Life (items 9a-k)	
		Family Decision Making (10a-d)	Steinberg
		Time Use (items 11-13)	
		Teacher-Child Rating Scale (Social Skills) (Items 14 & 20)	Hightower (adapted slightly by NCEDL for use by parents)
		Health (Items 15-16)	
End of Year Teacher Questionnaire	Late Spring 2003	IEP (Items 2 & 3)	NCEDL
		Absences (Item 5)	NCEDL

D. First Grade (2004) Instruments

		1st Grade Instrument	Author
Teacher Questionnaire	Spring of 1 st Grade (2004)	Teacher Demographics (Items 1 - 13)	NCEDL
		Class Characteristics (Items 14 – 25)	NCEDL
		Curriculum (Items 26 – 32)	NCEDL
		Working with Parents (Items 33 - 36)	NCEDL
		Assistant or Co-Teacher (Items 37 – 39)	NCEDL
Teacher Questionnaire about Study Child	Spring of 1 st Grade (2004)	Teacher-Child Rating Scale (Social Skills) (Items 6 & 7)	Hightower
		Severity of Mental Health Issues (Items 8 - 14)	Barbarin
		Special Services/Disability (Items 15 – 20)	NCEDL
		Academic Rating Scale: First Grade Version (Item 22 & 23)	ECLS-K
		Home-School Relationships (Items 24-30)	Barbarin
		Student-Teacher Relationship (Item 31)	Pianta
		Family Involvement (Item 32)	Barbarin

E. Data Collector Reliability on Observational Measures

Reliability: ECERS¹

Time Point	Type of test	Number of data collectors	Final Test Prior to Data Collection			
			Mean (sd): Weighted Kappa	Range: Weighted Kappa	Mean (sd): % within 1 ²	Range: % within 1
Fall of Pre-K (2001)	Live with one of two goldstandards ³	22	.65 (.07)	.53 to .83	83% (6.8)	71% to 97%
Spring of Pre-K (2002)	Live with one of two goldstandards or with data collection partner	23	.74 (.10)	.60 to .98	87% (6.1)	77% to 100%
Winter of Kindergarten (2002-2003)	Live with one of four goldstandards	24	.67 (.06)	.60 to .81	85% (5.7)	72% to 94%

¹ Prior to each wave of data collection, each data collector had to reach a weighted kappa of at least .60. (In the fall of pre-k, one data collector collected data despite being below that standard due to a calculation error.)

² “% within 1” refers to the percent of data collectors’ responses that were within one scale point of the gold standard’s response.

³ “Goldstandards” are full-time project staff who were highly trained and experienced in use of this measure. For the Snapshot and CLASS these individuals were authors of the measures.

Reliability: CLASS¹

Time Point	Type of test	Number of data collectors	Final Test Prior to Data Collection			
			Mean (sd): Weighted Kappa	Range: Weighted Kappa	Mean (sd): % within 1	Range: % within 1
Fall of Pre-K (2001)	Videotaped segments of pre-k classrooms	20	.67 (.13)	.49 to .90	89% (6.6)	80% to 100%
Spring of Pre-K (2002)	Live with one of four goldstandards	24	.73 (.11)	.42 to .84	93% (5.7)	80% to 100%
Fall of Kindergarten (2002)	Videotaped segments of kindergarten classrooms	24	.60 (.06)	.50 to .71	86% (4.8)	80% to 94%
Spring of Kindergarten (2002)	Videotaped segments of kindergarten classrooms or live visit with already reliable data collection partner	24	.68 (.07)	.51 to .82	87% (6.6)	80% to 98%

¹ Prior to each wave of data collection, each data collector had to reach a standard of 80% of responses within on scale point of the correct answer.

Reliability: Snapshot, Fall of Pre-K^{1, 2, 3}

Twenty-two data collectors tested using videotaped segments from pre-k classes.

	Mean (sd): Kappa	Range: Kappa
Overall (across all codes)	.67 (.04)	.59 to .73
Child Engagement (mean of medians)	.70 (.06)	.57 to .81
Read to	.76 (.21)	.00 to .92
Letter/sound	.75 (.16)	.31 to 1.00
Oral Language	.27 (.16)	.00 to .56
Writing	.61 (.17)	.32 to .85
Math	.54 (.11)	.31 to .79
Science	.65 (.11)	.48 to .82
Social Studies	.67 (.13)	.32 to .87
Aesthetics	.73 (.06)	.62 to .83
Gross Motor	.80 (.10)	.56 to .94
Fine Motor	.76 (.15)	.27 to .90
Adult Interaction ⁴		
Simple/Elaborated	.71 (.08)	.57 to .92
Teacher-Child Engagement (mean of medians)	.53 (.06)	.41 to .66
Encourage/Scaffolds ⁵	.62 (.08)	.42 to .79
Didactic	.44 (.13)	.21 to .71
Second Language	.62 (.31)	.00 to 1.00

¹ Prior to data collection, each data collector had to attain an overall kappa of at least .60 across all codes. Additionally, they had to attain a certain median kappa in each section. For Child Engagement and Adult Interaction the standard was .55; for Teacher Child Engagement it was .50.

² It was not possible to accurately determine the “activity setting” from the videotaped segments, so those codes were not tested prior to fall of pre-K data collection. However, they are very straightforward codes, and when data collectors were tested using live visits in the spring, their reliability was very high (see next table). Almost all of the data collectors were the same from fall to spring data collection, so we have every reason to believe the fall activity setting data are reliable.

³ Distracted and pre-reading appeared too few times in the videotapes to accurately assess reliability.

⁴ Data collectors could not achieve reliability on two of the “Adult Interaction” codes during this first wave of data collection. Those codes do not appear in the data set until Spring of Pre-k, when data collectors demonstrated reliability. Simple and Elaborated had to be combined during this first wave of data collection to attain sufficient data collector reliability. They appear as one code in the dataset.

⁵ Encourages and scaffolds had to be combined during this first wave of data collection to attain sufficient data collector reliability. They appear as one code in the dataset.

Reliability: Snapshot, Spring Pre-K^{1,2}

Twenty-three data collectors tested using live visits to pre-k classrooms with a goldstandard (one of the measure's authors). Because of the live visits, not all codes were seen enough times (minimum of 4) to count toward the reliability statistics (see the fourth column in the chart below).

Section/Item	Mean (sd): Kappa	Range: Kappa	Number of data collectors whose test included this code 4 or more times (out of 23)
Overall (across all codes)	.85 (.06)	.75 to .96	
Activity Setting (mean of medians)	.95 (.04)	.85 to 1.00	
Routine	.90 (.05)	.79 to 1.00	20
Meals/snacks	.99 (.03)	.90 to 1.00	10
Whole group time	.96 (.04)	.86 to 1.00	18
Free choice/center	.96 (.06)	.74 to 1.00	23
Individual time	.62 (.53)	.00 to .93	3
Small group time	.85 (.27)	.00 to 1.00	20
Outside time	.95 (.09)	.73 to 1.00	14
Child Engagement (mean of medians)	.82 (.10)	.65 to 1.00	
Read to	.91 (.11)	.57 to 1.00	19
Pre-read/read	.71 (.34)	.00 to 1.00	14
Letter/sound	.88 (.13)	.64 to 1.00	9
Oral Language	.63 (.30)	.06 to 1.00	19
Writing	.97 (.04)	.94 to 1.00	5
Math	.77 (.17)	.38 to 1.00	18
Science	.75 (.15)	.39 to 1.00	23
Social Studies	.76 (.13)	.43 to 1.00	23
Aesthetics	.85 (.09)	.64 to 1.00	21
Gross Motor	.88 (.12)	.68 to 1.00	14
Fine Motor	.79 (.17)	.46 to 1.00	20
Adult Interaction	.74 (.10)	.54 to .95	
Minimal	.67 (.15)	.42 to .82	7
Simple	.74 (.15)	.34 to .95	23
Elaborated	.77 (.10)	.66 to 1.00	9
Teacher-Child Engagement	.74 (.13)	.50 to .96	
Encourage	.81 (.14)	.48 to 1.00	21
Scaffolds	.68 (.19)	.12 to 1.00	17
Didactic	.71 (.14)	.49 to .93	23
Second Language	.94 (.08)	.89 to 1.00	2

¹ Prior to data collection, each data collector had to attain an overall kappa of at least .60 across all codes. Additionally, they had to attain a certain median kappa in each section. For Activity Setting, Child Engagement, and Adult Interaction the standard was .55; for Teacher Child Engagement it was .50.

² None of the live tests contained enough incidences of "Adult Interaction: Routine" to accurately assess reliability.

Reliability: Snapshot, Fall of Kindergarten^{1,2,3}

Twenty-four data collectors tested using videotaped segments from kindergarten classes.

	Mean (sd): Kappa	Range: Kappa
Overall (across all codes)	.77 (.01)	.76 to .78
Child Engagement (mean of medians)	.75 (.06)	.63 to .86
Read to	.90 (.07)	.72 to 1.00
Pre-read/read	.65 (.13)	.29 to .88
Letter/sound	.85 (.16)	.59 to .94
Oral Language	.64 (.16)	.39 to .91
Writing	.75 (.16)	.21 to 1.00
Copying	.65 (.11)	.42 to .79
Math	.78 (.11)	.58 to .90
Science ⁴	.38 (.27)	.00 to .88
Social Studies	.73 (.14)	.46 to .94
Aesthetics	.54 (.12)	.23 to .84
Fine Motor	.90 (.06)	.78 to 1.00
Adult Interaction	.87 (.09)	.80 to .93
Routine	.63 (.25)	.00 to 1.00
Minimal	.65 (.15)	.37 to .85
Simple	.66 (.16)	.31 to .93
Elaborated	.70 (.17)	.38 to 1.00
Teacher-Child Engagement	.60 (.08)	.50 to .76
Encourage	.70 (.09)	.51 to .90
Scaffolds	.45 (.13)	.10 to .67
Didactic	.61 (.11)	.41 to .80

¹ Prior to data collection, each data collector had to attain an overall kappa of at least .60 across all codes. Additionally, they had to attain a certain median kappa in each section: Child Engagement, Adult Interaction, Teacher-Child Engagement. For Child Engagement and Adult Interaction the standard was .55; for Teacher Child Engagement it was .50.

² It was not possible to accurately determine the “activity setting” from the videotaped segments, so those codes were not tested prior to fall of Kindergarten data collection. However, they are very straightforward codes and when data collectors were tested using live visits during the prior spring, their reliability was very high (see last table). Almost all of the data collectors were the same across the two years of the project, so we have every reason to believe the fall activity setting data are reliable.

³ Distracted, Gross Motor, and 2nd Language did not appear on the video tapes too few times (<3) for meaningful scores to be calculated.

⁴ Only 4 instances of Science occurred in this test. This exceeded the cut-off of 3 or more instances so it is included here, but the values should be interpreted with caution.

Reliability: Snapshot, Spring of Kindergarten^{1,2,3}

Twenty-three data collectors were tested using video-taped segments from kindergarten classes (n = 20) or a live visit with an already reliable data collection partner (n=3).

	Mean (sd): Kappa	Range: Kappa
Overall (across all codes)	.74 (.04)	.68 to .81
Child Engagement	.77 (.06)	.63 to .88
Read to	.86 (.13)	.64 to 1.00
Pre-read/read	.62 (.26)	.00 to 1.00
Letter/sound	.84 (.06)	.70 to .95
Oral Language	.61 (.17)	.32 to 1.00
Writing	.86 (.15)	.39 to 1.00
Math	.77 (.08)	.58 to .87
Science	.44 (.26)	.00 to .89
Social Studies	.81 (.07)	.63 to .90
Aesthetics	.75 (.15)	.44 to .93
Fine Motor	.68 (.15)	.47 to 1.00
Adult Interaction	.72 (.10)	.55 to 1.00
Routine	.80 (.13)	.55 to 1.00
Minimal	.63 (.23)	.00 to 1.00
Simple	.64 (.15)	.31 to 1.00
Elaborated	.71 (.19)	.22 to 1.00
Teacher-Child Engagement	.61 (.07)	.51 to .81
Encourage	.66 (.09)	.52 to .90
Scaffolds	.54 (.14)	.29 to .90
Didactic	.64 (.08)	.46 to .81
Second Language	.86 (.35)	.00 to 1.00

¹ Prior to data collection, each data collector had to attain an overall kappa of at least .60 across all codes. Additionally, they had to attain a certain median kappa in each section. For Child Engagement and Adult Interaction the standard was .55; for Teacher Child Engagement it was .50.

² It was not possible to accurately determine the “activity setting” from the videotaped segments, so those codes were not tested prior to fall of pre-K data collection. However, during training, the measure’s authors tested the data collectors on their use of these codes. Each data collector correctly identified the activity setting over 80% of the time.

³ Distracted, copying, and gross motor appeared too few times in the videotapes to be counted as a valid test.

ANALYSIS VARIABLES **(SUMMARY SCORES)**

Classroom and Teacher assessments.

Appendices B, C, & D of the Study Description and Documentation outline all measures collected at each time point. Observed classroom quality and practice were measured using three measures: two that rated global quality (ECERS-R, CLASS) and one that assessed children’s activities and teacher practices (Snapshot). These instruments are described below. During the pre-k year, ECERS-R observations were conducted once in the fall and once in the spring; CLASS/Snapshot observations were conducted for two days in the fall and two days in the spring. During the kindergarten year, we sought to conduct Snapshot and CLASS observations three times during the year and the ECERS-R once in the winter. Snapshot and CLASS data in this dataset have been combined across all observations for the kindergarten year. (See Appendix E of the Study Description and Documentation for a summary of the data collector reliability testing and statistics for all observational measures.) Teachers were asked to complete questionnaires about themselves, including information on demographic characteristics (e.g., race, age, education, training), beliefs about children, and depressive symptoms. Teachers completed questionnaires in the fall and spring of the pre-kindergarten and kindergarten years and again in the spring of the 1st grade year. Program administrators (e.g., principals, directors) completed a questionnaire in the spring of the pre-kindergarten year.

Early Childhood Environment Rating Scale-Revised. Global classroom quality was assessed using the ECERS-R (Harms & Clifford, 1983; Harms, Clifford, & Cryer, 1998). The ECERS-R is specifically designed for use in classrooms serving children 2.5 -5 years of age. Scores on the ECERS-R range from 1-7 with 1 indicating “inadequate” quality, 3 indicating “minimal” quality, 5 indicating “good” quality, and 7 indicating “excellent” quality. See “ECERS-R Additional Notes – PK” and “ECERS-R Additional Notes – Kindergarten” for study-specific scoring information. In order to decrease the burden on teachers, this study did not include the “Parent and Staff” items (38 through 43) which require extensive additional questioning of the teacher. Further, the “Toileting” item (number 12) is not include in any of the scoring because this study used stricter rules for scoring this item than past research. The ECERS-R observation was completed by a different data collector from the individual who conducted the Snapshot and CLASS observations during the pre-k year. During the Kindergarten year, it was generally done by the same individual as the Snapshot/CLASS, on a different day.

On each ECERS-R pre-k observation day, data collectors stayed from the beginning of the class until the end of the class in part-day rooms, or from the beginning of class until children were settled into nap in full-day rooms. On each ECERS-R kindergarten observation day they stayed for the entire class in both full- and part-day rooms, taking a thirty minute break for lunch at a time that did not interfere with their coding.

In addition to the overall Composite Score, factor analysis of the ECERS-R yielded two factors. The first factor, termed Teaching and Interactions, consisted of loadings from 11 individual items: Greeting/Departing, Encouraging Children to Communicate, Using Language to Develop Reasoning Skills, Informal Use of Language, Supervision of Gross Motor Activities, General Supervision of Children, Discipline, Staff-Child Interactions, Interactions among Children, Free Play and Group Time. The second factor, labeled Provisions for Learning, consisted of loadings from 12 items: Room Arrangement, Space for Privacy, Gross Motor Equipment, Fine Motor, Art, Blocks, Sand/Water, Dramatic Play, Nature/Science, Schedule, Free Play, and Group Time.

Classroom Assessment Scoring System (CLASS) (Pianta, La Paro, & Hamre, 2004). The CLASS provides an assessment of the classroom in terms of quality of emotional climate, classroom management, and instructional supports for learning. During pre-k, the observer rated the pre-kindergarten classroom and the teacher on 9 dimensions roughly every 30 minutes, throughout the two observation days in the fall and two observation days in the spring (same days as the Snapshot). During kindergarten, the CLASS was again conducted roughly every 30 minutes during the observation days (again, the same days as the Snapshot). The goal was to conduct three separate CLASS/Snapshot observation days during the kindergarten year, spread throughout the school year¹. In pre-k, each classroom's score is the average of its scores across the two observation days, so there is one fall value and one spring value. In kindergarten, each classroom's score is the average of its scores across all observations days, so there is one kindergarten value.

Each of nine dimensions is rated from 1-7 with 1 or 2 indicating the classroom is low on that dimension, and 3, 4, or 5 indicating that the classroom is in the mid-range, and a 6 or 7 indicating the classroom is high on that dimension. Positive Climate reflects the enthusiasm, enjoyment, and respect displayed during interactions between the teacher and children and among children. Negative Climate is the degree to which the classroom has a negative emotional and social tone (displays of anger, aggression, and/or harshness). Teacher Sensitivity is the extent to which teachers provide comfort, reassurance, and encouragement. Over-control reflects the extent to which classroom activities are rigidly structure or regimented. Effective Behavior Management encompasses the teacher's ability to use effective methods to prevent and redirect children's misbehaviors. Productivity reflects how well the teacher manages instructional time and routines so that children learn and make progress. Concept Development considers the strategies teachers employ to promote children's higher order thinking skills and creativity through problem-solving, integration, and instructional discussions. Instructional Learning Formats includes the available activities, method of presentation, use of groupings, and range of materials that teachers use to maximize children's engagement. Finally, Quality of Feedback focuses on the quality of verbal evaluation provided to children about their work, comments, and ideas. Contact the measure's authors for more detailed information on scoring.

¹ The actual number of Snapshot/CLASS observations days during kindergarten varied by child, depending on absences. The variable SNAPDAYK indicates the number of kindergarten Snapshot/CLASS observation days for each child.

Factor analysis of the CLASS yielded two factors. Emotional Climate is a composite of Positive Climate, Negative Climate (reversed), Teacher Sensitivity, Over-control (reversed), and Behavior Management. The second factor, termed Instructional Climate, is a composite of Concept Development and Quality of Feedback.

Emerging Academics Snapshot. We used the Emerging Academics Snapshot (Ritchie, Howes, Kraft-Sayre, & Weiser, 2001) to describe children's experiences within their program. The Snapshot consists of 27 items that are coded as present or absent within a twenty-second period. The items are divided into sections including children's activity setting, children's engagement, and children's interaction with adults. See "Snapshot Codebook – PK FINAL" and "Snapshot Codebook – Kindergarten FINAL" for descriptions of each code.

Each Snapshot observation consists of a 20-second observation period, followed by a 40-second coding period. Data collectors observed each of the study children in the classroom succession, with breaks about every 30 minutes to code the CLASS. The average number of observations at each time point was: 51.1 times ($sd = 18.7$, $range = 8$ to 120) in the fall of the pre-k year over a two-day period; 65.5 ($sd = 23.3$, $range = 16$ to 159) in the spring of the pre-k year over a two-day period; and 381.4 ($sd = 144.0$, $range = 38$ -778) over a three day period². The dataset contains variables indicating the number of observations and number of observation days for each child at each time point (SNAPOBSPF, SNAPDAYPF, SNAPOBSPS, SNAPDAYPS, SNAPOBSK, SNAPDAYK).

On each Snapshot/CLASS pre-k observation day, data collectors stayed from the beginning of the class until the end of the class in part-day rooms, or from the beginning of class until children were settled into nap in full-day rooms. On each Snapshot/CLASS kindergarten observation day they stayed for the entire class in both full- and part-day rooms, but did not observe during lunch, recess directly following lunch, or nap. **Because the observation periods were systematically different, the pre-k and kindergarten Snapshots should not be directly compared (especially with regard to meals/snacks, recess, outside time and gross motor).**

Data were summarized in two ways. First, the mean was computed across all observations in a classroom to represent practices at the classroom level, and the descriptive statistics for that level of summarization are shown below in Table 1. Second, the mean was computed across all observations of a given child to represent that child's experiences. The summary statistics for those measures are shown in Table 2. Note that many children changed classrooms during the kindergarten year, so the child-level kindergarten values sometimes represent observations in multiple classrooms for the same child. The variable SNAPCLASSK indicates the number of different classrooms in which each child was observed for the Snapshot.

² During the pre-kindergarten year, almost all study-classrooms contained 4 study-children and the observation time was split among them. Most kindergarten classrooms contained fewer study-children, with many classrooms containing only 1 study-child. Therefore, the kindergarten observation days were split among significantly fewer children, thereby increasing the number of observations per child on each study day.

Teacher traditional attitudes. Teachers' beliefs were measured with the Modernity Scale (Schaefer & Edgerton, 1985), a 16-item Likert-type questionnaire that discriminates between "traditional" or relatively authoritarian approaches to child rearing and more "modern or progressive" child-centered approaches. Scores are derived by taking the mean of all items, with non-traditional beliefs reversed-scored. Teachers holding a more traditional view agreed with statements such as "Children must be carefully trained early in life or their natural impulses make them unmanageable" and "Children should always obey the teacher." Teachers with more progressive beliefs agreed with statements such as "Children should be allowed to disagree with their parents if they feel their won ideas are better." In the current dataset, teachers with higher scores have more progressive beliefs. Cronbach's alpha for this scale was reported as .89 in one study (NICHD ECCRN, 2002), as .84 by the scale's authors and was .78 in the pre-k teacher sample.

Teachers' depressive feelings. The level of depression experienced by the teacher was assessed using the Center for Epidemiological Studies Depression Scale (CES-D; Radloff, 1977), a 20-item questionnaire that identifies depressive symptomatology in the general population. Scores are created by taking the mean of all items, with positive items reversed-scored so that higher scores on this scale reflect more emotional distress. Coefficient alpha for this scale has been reported as .82 (NICHD ECCRN, 2002) and was .86 in the present sample of pre-k teachers. Test re-test reliability is reported as .57 (Radloff, 1977).

Self-efficacy. Pre-K and kindergarten teacher's perceptions of her own self-efficacy were assessed using a version of Bandura's Teacher Self-Efficacy scale that had been modified for the NCEDL Kindergarten Transition Project. The scale consists of the mean of 10 items that are rated on a 1 (nothing) to 5 (a great deal) scale. Sample items include: How much can you do to get through to the most difficult students? and How much can you do to increase students' memory of what they have been taught in previous lessons?

Classroom and program attributes. Questionnaires to pre-k program directors and to teachers provided information on whether the classroom was *located in a school building* or not, the hours the program operated (which is included as continuous variables [HOURSDAYP, HOURSDAYK] and dichotomous variables [FULLDAYP and FULLDAYK] where full day is more than 20/week) and *child-teacher ratio* (number of children/number of staff). Teachers also provided information on the number of *years of education* they had attained, their *years of teaching experience with 4 year-olds, kindergarteners, older children and younger children*, their *ethnicity*, and *teaching credentials* (certificates, endorsements, etc.). In addition, during the course of selecting the sample of children, we sent very brief demographic questionnaires home to all children in the selected pre-k classrooms. One question assessed the family's income level, which we then categorized and above or below 150% of the poverty line for the family's size. The current data set has a variable for each child indicating if his/her family is above or below 150% of the poverty line and a variable for each pre-k classroom indicating the percent of children who are below 150% of poverty.

CLASSROOM OBSERVATIONS (class-level, one observation per classroom)

CHILD ADULT RATIO	Fall Pre-K (n = 227)		Spring Pre-K (n = 240)		K (n = 869)		Corr. Pre-K Fall to Spring	Variable Name
	Mean	SD	Mean	SD	Mean	SD	r	
Class size (observed)	16.0	4.1	16.0	3.8	18.63	3.76	0.66	csizeobpf, csizeobps, csizeobk
Class size (enrolled)	18.3	4.2	18.7	3.9	20.54	3.72	0.77	csizeenpf, csizeenps, csizeenk
Number of adults observed	2.4	0.7	2.5	0.8	1.72	0.63	0.69	staffobpf, staffobps, staffobk
Ratio:					(n = 866)			
# children observed / adults observed	6.9	2.1	7.0	2.1	12.30	4.95	0.63	ratioobpf, ratioobps, ratioobk
Ratio:					(n = 865)			
# children enrolled / adults observed	7.9	2.2	8.1	2.1	13.60	5.44	0.68	ratioenpf, ratioenps, ratioenk

GLOBAL QUALITY	Fall Pre-K (n = 227)			Spring Pre-K (n = 240)			K (n = 869)			Corr. Pre-K Fall to Spring	Variable Name
	Mean	SD	α	Mean	SD	α	Mean	SD	α	r	
ECERS-R total	3.77	0.82	0.90	3.76	0.80	0.90	3.18	0.59	0.87	0.69	ecertotpf, ecertotps, ecertotk
2 Factors											
<u>Factor 1: Language & Interactions</u> (Items 9,16,17,18,29,30,31,32,33,35,36)	4.43	1.29	0.88	4.44	1.22	0.88	4.21	1.05	0.84	0.60	ecerf1pf, ecerf1ps, ecerf1k
<u>Factor 2: Activities & Schedule</u> (Items 3,4,8,19,20,22,23,24,25,34,35,36)	3.79	0.96	0.83	3.79	0.88	0.83	2.67	0.65	0.79	0.72	ecerf2pf, ecerf2ps, ecerf2k
Correlations between factors	r = .61			r = .78			r = .63				
% variance accounted for	31%			31%							

Note: confirmatory factor analysis of the spring data supported the two factor solution from the fall data (RMSEA = .059 — guidelines RMSEA < .08 is good fit)

CLASSROOM CLIMATE	Fall Pre-K (n = 227)			Spring Pre-K (n = 240)			K (n = 869)			Corr. Pre-K Fall to Spring	Variable Name
	Mean	SD	α	Mean	SD	α	Mean	SD	α	r	
CLASS total	4.39	0.75	0.89	4.34	0.68	0.89	4.49	0.54	0.86	0.62	clastotpf, clastotps, clastotk

2 Factors												
<u>Factor 1: Emotional Climate</u> items: positive climate, negative climate, teacher sensitivity, over control, behavior management		5.22	0.76	0.85	5.34	0.75	0.86	5.54	0.61	0.85	0.59	clasf1pf, clasf1ps, clasf1k
<u>Factor 2: Instructural Climate</u> items: concept development, quality of feedback		2.47	1.10	0.87	1.94	0.82	0.85	1.98	0.67	0.85	0.32	clasf2pf, clasf2ps, clasf2k
Correlations between factors		r = .41			r =.34			r = .30				

	SNAPSHOT– Pre-K Class Level (SNAPSHOT was collected only at child level at K)					
	Fall Pre-K (n = 231)		Spring Pre-K (n = 240)		Correlation of Fall to Spring	Variable Name
Activity Setting	Mean	SD	Mean	SD	r	
Free choice / center	0.35	0.15	0.33	0.14	0.68	asfcc_pf, asfcc_ps
Routine	0.20	0.08	0.21	0.07	0.42	asrtn_pf, asrtn_ps
Snacks / meals	0.13	0.07	0.14	0.06	0.57	asmls_pf, asmls_ps
Small group	0.07	0.09	0.06	0.08	0.45	assgt_pf, assgt_ps
Whole group	0.24	0.12	0.23	0.10	0.59	aswgt_pf, aswgt_ps
Individual	0.02	0.04	0.04	0.04	0.38	asint_pf, asint_ps
Adult Interactions	Mean	SD	Mean	SD	r	Variable Name
Elaborated	0.06	0.05	0.04	0.03	0.35	aielbc_pf, aielbc_ps
Routine	0.01	0.01	0.01	0.01	0.05	airtn_pf, airtn_ps
Simple	0.06	0.04	0.07	0.03	0.40	aismpc_pf, aismpc_ps
Minimal	0.04	0.03	0.03	0.02	0.20	aiminc_pf, aiminc_ps
Child Engagement	Mean	SD	Mean	SD	r	Variable Name
Read to	0.03	0.03	0.03	0.03	0.03	ceprrc_pf, ceprrc_ps
Letter / Sound	0.03	0.04	0.03	0.03	0.44	celtsc_pf, celtsc_ps
Oral Language	0.06	0.05	0.05	0.05	0.14	ceoldc_pf, ceoldc_ps
Math	0.07	0.05	0.07	0.04	0.32	cemthc_pf, cemthc_ps
Science	0.10	0.06	0.10	0.07	0.30	cescic_pf, cescic_ps
Writing	0.01	0.01	0.01	0.01	0.09	cewrtc_pf, cewrtc_ps
Social Studies	0.19	0.09	0.18	0.08	0.29	cescsc_pf, cescsc_ps
Adult Interactions (Mutually exclusive)	Mean	SD	Mean	SD	r	Variable Name
Elaborated	0.10	0.06	0.08	0.05	0.33	aielbc_c_pf, aielbc_c_ps
Routine	0.01	0.01	0.01	0.01	0.05	airtn_c_pf, airtn_c_ps
Simple	0.20	0.09	0.18	0.07	0.44	aismpc_c_pf, aismpc_c_ps
Minimal	0.69	0.11	0.72	0.09	0.49	ainonec_c_pf, ainonec_c_ps
Child Engagement (Mutually exclusive)	Mean	SD	Mean	SD	r	Variable Name
Read to	0.06	0.04	0.05	0.04	0.24	cereadc_c_pf, cereadc_c_ps
Letter / Sound	0.03	0.04	0.03	0.03	0.44	celtsc_c_pf, celtsc_c_ps
Oral Language	0.06	0.05	0.05	0.04	0.14	ceoldc_c_pf, ceoldc_c_ps
Math	0.05	0.04	0.05	0.04	0.35	cemthc_c_pf, cemthc_c_ps
Science	0.07	0.05	0.07	0.05	0.33	cescic_c_pf, cescic_c_ps
Writing	0.01	0.01	0.01	0.01	0.14	cewrtc_c_pf, cewrtc_c_ps
Social Studies	0.13	0.07	0.12	0.06	0.24	cescsc_c_pf, cescsc_c_ps
Teacher-Child Engagement	Mean	SD	Mean	SD	r	Variable Name
Encourage	0.17	0.1	0.14	0.08	0.35	tcencc_pf, tcencc_ps
Scaffolds	0.05	0.04	0.05	0.04	0.28	tcscfc_pf, tcscfc_ps
Didactic	0.16	0.16	0.23	0.1	0.48	tcddtc_pf, tcddtc_ps
Second language	0.02	0.05	0.02	0.04	0.83	tcsccl_pf, tcsccl_ps

Data collectors indicated all child engagement activities seen during the 20-second observations (check all that apply, not mutually exclusive). Mutually exclusive child engagement variables have been created for analysis purposes. Both the mutually exclusive and the non-mutually exclusive variables are included in the data set. The mutually exclusive variables were created in the following manner:

- If letter word indicated it is counted,
- otherwise if “oral language” is indicated then it is counted
- otherwise if “read to” is indicated then it is counted
- otherwise if “math” is indicated then it is counted
- otherwise the following are counted in this order – science, writing, social studies, aesthetics, none

Correlation (fall of pre-k): whole group and free choice/centers $r = -.59$, other correlations among activity settings, interaction, and engagement codes were less than .30.

TEACHER QUESTIONNAIRES (Pre-K - class level)

	Fall Pre-K				Variable Name
	N	Mean	SD	α	
T. experience	233	13.17	9.35		exptotp
Certification	222	72%			cert4yrp
CESD	236	8.61	7.48	0.86	t_deppf
Ideas about childrearing	233	38.9	9.88	0.78	t_ideapf
Self-efficacy	231	4.06	0.49	0.86	t_beliefpf
Teacher has BA or more	237	59%			t_educp
CDA	218	32%			t_cdap
Length of day (hours)	231	5.09	2.47		schdayp
Class is full day	230	48%			Fulldayp
Enrolled child/adult ratio	226	7.92	2.25		Ratioenpf
Class size	237	18.15	3.51		cls_sizepf
Number of teachers/adults	226	2.44	0.7		Staffobpf

TEACHER QUESTIONNAIRES (K and G1 - child level)

	Kindergarten				Grade 1				Variable Name
	N	Mean	SD	α	N	Mean	SD	α	
T. experience	857	14.42	9.97		589	13.88	9.93		exptotk, exptote
Certification	839	96%			574	0.97			bacert_ck, bacert_de
CESD	816	6.83	6.69	0.76					t_cesdk
Ideas about childrearing	837	39.46	9.14	0.76					t_ideak
Self-efficacy	822	4.04	0.48						tchsleffk
Teacher has BA or more	870	99%			593	1.00			t_educk
CDA	793	17%							t_cdak
Length of day (hours)	867	5.52	1.65						schdayk
Class is full day	867	75%							fulldayk
Enrolled child/adult ratio	865	13.6	5.44						ratioenk
Class size	874	20.42	3.81		588	20.41	3.93		cls_sizekf, cls_sizee
Hours per day	870	5.52	1.65						hoursdayk
Number of teachers/adults	866	1.72	0.63						staffobk

Teacher race – coded as 6 categories: 1.African American if only the category of Black/African American indicated; 2. Asian/Pacific Islander if one or more of the following categories indicated: Pacific Islander, Chinese, Filipino, Japanese, Korean, Vietnamese, Asian Indian, and Other Asian; 3. White if only the category of White/Caucasian indicated; 4. Hispanic if one or more of the following

categories indicated: Mexican American/Chicano, Puerto Rican, Cuban, and Other Hispanic/Spanish/Latino; 5. Native American/Indian if only the category of Native American/Indian indicated; 6. Mixed if not included in any of the above 5 categories except when the category of Other indicated and specified (then recoded to appropriate category).

ADMINISTRATOR QUESTIONNAIRE

	Spring Pre-K	
	N	Mean
Location - in school?	245	54%
Head start?	238	30%

Individual Child Assessment

Children were directly assessed in the fall and spring of their pre-kindergarten year and again in the fall and spring of the kindergarten year. Additionally, teachers completed questionnaires about children's academic and social skills in the fall and spring of the pre-kindergarten year, the fall and spring of the kindergarten year, and the spring of the 1st grade year. We used a multi-method approach to assessment. We individually assessed children using standardized measures of academic achievement, e.g., The Woodcock-Johnson Tests of Achievement, (Woodcock, McGrew, & Mather, 2001), as well as directly measuring skills considered by policy makers to indicate school success (e.g., counting and naming letters). A trained data collector assessed each participating child. On average, these assessments were 151.3 days apart from fall of pre-k to spring of pre-k ($SD = 25.0$, $range = 91 - 244$), 203.1 ($SD = 40.0$, $range = 103 - 311$) days apart from spring of pre-k to fall of k, and 185.3 ($SD = 32.1$, $Range = 99 - 265$) days from fall of k to spring of k. The entire battery took between 45 minutes and one hour and was conducted during the school day, outside the classroom. In order to ensure that the child assessment battery was being correctly administered, data collectors submitted videotapes of themselves conducting the assessment or conducted an assessment that was observed by one of the study supervisors. Additionally we asked teachers to rate children on scales indicating academic achievement, e.g., ECLS-K Academic Rating Scales (U. S. Department of Education & Education Statistics, 1999) and social and motivational skills (Hightower, Work, Cowen, Lotyczewski, Spinell, Guare, & Rohrbeck, 1986).

preLAS2000 (Duncan & De Avila, 1998). Children who spoke a language other than English at home were given a portion of the preLAS2000 at the beginning of the direct assessment in the fall of pre-k to screen for English proficiency. Three of the five scales of the preLAS (Simon Says, Art Show, and Human Body) were given in English. These scales require children to respond to a set of instructions (listening comprehension) and verbally identify pictures (receptive vocabulary). Children who scored 31 or more (out of 40) were given the English assessment battery. Children who scored 30 or below and spoke Spanish at home were given the Spanish assessment battery. Children who scored 30 or below and spoke a language other than Spanish at home, were not assessed at that time, but were tested in English during subsequent assessments. This system for determining assessment language was similar to the OLDS used by the ECLS-K (U. S. Department of Education & Education Statistics, 1999).

According to the Examiner's Manual, Cronbach's alphas were .88/.90 (Simon Says), .90/.89 (Art Show) and .86/.86 (Human Body). Test-retest reliabilities were .89 (Simon Says), .94 (Art Show) and .91 (Human Body).

At subsequent assessments points (spring of pre-k, fall of K, and spring of K), only children who had taken the preLAS at the previous assessment and not passed were given the preLAS again. At each time point, children who did not pass and spoke Spanish at home were given a Spanish language battery. Children who did not take the preLAS or who passed the preLAS, were given the English-language battery. Thus, at each assessment time point, the number and percent of children who took the Spanish-language battery decreased.

English-Language Assessment Battery

Peabody Picture Vocabulary Test 3rd edition (PPVT-III) (Dunn & Dunn, 1997). The PPVT-III is a test of receptive vocabulary and has shown to relate other measures of language, literacy, and academic achievement (Chow & McBride-Chang, 2003; Dunn & Dunn, 1997). Children are shown a set of 4 pictures and are asked to select the picture that best represents the meaning of a word spoken by the examiner. A standard score is computed for this scale. According to the scale's authors, the alpha coefficient of all the items on this scale range from .92 to .98 with a median reliability of .94.

Oral & Written Language Scale (OWLS) (Oral Expression Scale) (Carrow-Woolfolk, 1995). The Oral Expression Scale is a standardized measure designed to assess the understanding and use of spoken language. The measure's authors report correlations between the OWLS and achievement tests that range from .44 to .89. During the assessment, the examiner reads a verbal stimulus aloud while the child looks at a stimulus board containing one or more pictures. Children are required to respond orally by answering a question, completing a sentence, or generating a new sentence (or sentences). A standard score is computed on this scale. According to the measure's authors, test-retest reliability for the 4-to 5-year-old age range on the Oral Expression Scale is .86.

Woodcock-Johnson III Tests of Achievement: Applied Problems Subtest (Woodcock et al., 2001)). The Woodcock-Johnson III Tests of Achievement are well-established measures of academic achievement. The Applied Problems subtest of this standardized measure examines the ability to analyze and solve math problems. The reliability coefficients for the 3- to 5-year-old age group range from .92 to .94, according to the measure's authors.

Woodcock-Johnson III Tests of Achievement: Sound Awareness-Rhyming Subtest (Woodcock et al., 2001). The Sound Awareness-Rhyming Subtest measures the ability to rhyme, a sub-category of phonological awareness. The maximum possible on this scale is 17. The reliability coefficients for the 4- to 5-year-old age group range from .71 to .85 according to the measures' authors.

Woodcock-Johnson III Test of Achievement: Letter Word Identification (Woodcock et al., 2001) (administered during the kindergarten year only): This subtest measures word identification skills. The child is initially asked to identify letters. The remaining items require the child to pronounce written words correctly. The reliability coefficients for the 3- to 5-year-old age group range from .97 to .99 according to the measure's authors.

Identifying Letters. The ability to identify letters is an indicator of emergent literacy. In this assessment, children are shown a set of mixed capital and lowercase letters and asked to identify as many letters as they could. The highest possible score is 26.

Identifying Numbers. The ability to identify numbers is an indicator of emergent numeracy. Children are shown a sheet of numbers (1-10 printed in random order) and asked to identify as many numbers as they could. The maximum possible score is 10.

Counting. This is a criterion-referenced measure of the ability to count objects. The ability to identify numbers is an indicator of emergent numeracy (Gelman & Gallistel, 1986). In this task, children were asked to count and point, with a one-to-one correspondence, using a picture card with 20 teddy bears. If a child counted to 20 correctly, another sheet of bears was immediately presented for continued counting. The highest number the child counts in sequence is recorded, with a maximum score of 40.

Color Bears (Head Start Family and Child Experiences Survey, 1998). A criterion-referenced measure, Color Bears measures the ability to name 10 basic colors. To assess color recognition and identification, children were shown a page of 10 different colored bears and asked which colors they can name. They pointed to the bear as they named the color. The maximum score is 10.

Spanish-Language Assessment Battery

Batería Woodcock-Muñoz-Revisada: Pruebas de Aprovechamiento, Problemas Aplicados (Woodcock & Muñoz-Sandoval, 1996). The Problemas Aplicados subtest of the Woodcock Muñoz examines the ability to analyze and solve math problems. According to the Examiner's Manual the internal consistency reliability coefficient (R_{11}) is .908 for four-year-olds and .841 for six-year-olds. The standard error of measurement is 6.5 for 4-year-olds and 5.4 for 6-year-olds.

Batería Woodcock-Muñoz-Revisada: Pruebas de Aprovechamiento, Indentificación de Letras y Palabras (Woodcock & Muñoz-Sandoval, 1996) (administered during the kindergarten year only). The Indentificación de Letras y Palabras subtest examines reading identification skills in identifying isolated letters and words that appear in large type. The child is not required to know the meaning of any word. According to the Examiner's Manual the internal consistency reliability coefficient (R_{11}) is .916 for four-year-olds and .964 for six-year-olds. The standard error of measurement is 7.0 for four-year-olds and 5.0 for six-year-olds.

Test de Vocabulario en Imagenes Peabody (TVIP) (Dunn, Lugo, Padilla, & Dunn, 1986). The TVIP is an achievement test of receptive vocabulary. Children are shown a set of 4 pictures and are asked to select the picture that best represents the meaning of a word spoken by the examiner. According to the Examiner's Manual, the split-half reliability coefficient is .94 for 4-year-olds and .93 for 5-year-olds. The standard error of measurement for the standard scores was 4 at both ages. Note, starting in kindergarten, children given the Spanish battery were also given the PPVT in English to track progress in their English skills. PPVT scores attained as part of the Spanish battery are separate variables in the data set from those attained as part of the English battery.

Naming Letters, Numbers, Colors, and Counting: The Spanish battery included Spanish language versions of these same tests of basic skills. Spanish and English scores are separate variables in the data set.

Teacher Ratings

Academic Rating Scale: Language and literacy skills (NCES, 1999). Teachers rated study-children's language and literacy skills in the fall and the spring of pre-k, the fall and spring of kindergarten and the spring of first. The ratings were meant to cover a broad range of skills that children in pre-kindergarten and early elementary school might show. Thus, some of the skills listed may not be appropriate for all ages of young children, but were included to assess how children's skills change over time. The items were rated on a 1 to 5 scale with 1 = Not Yet, 2 = Beginning, 3 = In Progress, 4 = Intermediate, and 5 = Proficient. (Responses of N/A were recoded to equal 1) See the questionnaires for items. Note: the 1st Grade version of this instrument was used in 1st Grade. Reliability of this measure for the study sample was high ($\alpha=.89$ at pre-K fall, $\alpha=.92$ at pre-K spring, $\alpha=.92$ at K fall, and $\alpha=.94$ at K spring).

Academic Rating Scale: Mathematical Reasoning (NCES, 1999). Teachers rated study-children's mathematical reasoning skills the spring of pre-k, fall and spring of K, and spring of first grade. The ratings were meant to cover a broad range of skills that children in pre-kindergarten and early elementary school might show.. Thus, some of the skills listed may not be appropriate for all ages of young children, but were included to assess how children's skills change over time. The items were rated on a 1 to 5 scale with 1 = Not Yet, 2 = Beginning, 3 = In Progress, 4 = Intermediate, and 5= Proficient. (Responses of N/A were recoded to equal 1) See the questionnaires for items. Note: the 1st Grade version of this instrument, containing different items, was used in 1st Grade. Alpha for this sample was .94 at pre-K spring, K- fall and K-spring and .95 at 1st Grade.

Social skills and behavior problems. Teachers also completed questions concerning study children's social skills and behavior problems (Hightower et al., 1986) at each time point (fall and spring of pre-k, fall and spring of K, spring of 1st grade). Teachers scored children individually using a scale from 1-5 on how well statements described the child. For the Social Skills items, the scale was 1 = Not at all, 2=A Little, 3 = Moderately well, 4=Well, and 5 = Very well. For the Behavior Problem items, the response scale was 1 = Not at problem, 2=Mild, 3 = Moderate, 4=Serious, and 5 = Very serious problem. The reliability coefficients for Social Skills were .95 at pre-K fall and pre-K spring, .94 at K-fall, .95 at K-spring and .96 at 1st Grade. The internal consistency among the Behavior Problem items were also high at $\alpha=.92$ from pre-K fall through K-spring and at $\alpha=.93$ at 1st Grade.

Following instructions from the scale authors, 2 scales (competence and problems) and 7 subscales were created (assertiveness, peer social skills, task orientation, conduct problems, frustration tolerance, internalizing, learning problems). The tables below indicate which items are included with which scales and subscales.

Item	Scale	Subscale
Defends own views under group pressure	Competence	Assertiveness
Comfortable as a leader	Competence	Assertiveness
Participates in class discussions	Competence	Assertiveness
Expresses ideas willingly	Competence	Assertiveness
Questions rules that seem unfair/unclear	Competence	Assertiveness
Has many friends	Competence	Peer Social Skills
Is friendly toward peers	Competence	Peer Social Skills
Makes friends easily	Competence	Peer Social Skills
Classmates wish to sit near this child	Competence	Peer Social Skills
Well liked by classmates	Competence	Peer Social Skills
Works well without adult support	Competence	Task Orientation
Functions well even with distractions	Competence	Task Orientation
Completes work	Competence	Task Orientation
Well organized	Competence	Task Orientation
A self starter	Competence	Task Orientation
Accepts things not going his/her way	Competence	Frustration Tolerance
Ignores teasing	Competence	Frustration Tolerance
Accepts imposed limits	Competence	Frustration Tolerance
Copes well with failure	Competence	Frustration Tolerance
Tolerates frustration	Competence	Frustration Tolerance
Disruptive in class	Problems	Conduct Problems

Item	Scale	Subscale
Fidgety, difficulty sitting still	Problems	Conduct Problems
Disturbs others while they are working	Problems	Conduct Problems
Constantly seeks attention	Problems	Conduct Problems
Overly aggressive to peers (fights)	Problems	Conduct Problems
Defiant, obstinate, stubborn	Problems	Conduct Problems
Withdrawn	Problems	Internalizing Problems
Shy, timid	Problems	Internalizing Problems
Anxious, worried	Problems	Internalizing Problems
Nervous, frightened, tense	Problems	Internalizing Problems
Does not express feelings	Problems	Internalizing Problems
Unhappy, sad	Problems	Internalizing Problems
Underachieving (not working to ability)	Problems	Learning Problems
Poor work habits	Problems	Learning Problems
Poor concentration, limited attention span	Problems	Learning Problems
Difficulty following directions	Problems	Learning Problems
Poorly motivated to achieve	Problems	Learning Problems
Learning academic subjects	Problems	Learning Problems

Student-Teacher Relationships Scale (STRS; Pianta, 2001). In the spring of each each year (pre-k, kindergarten, 1st grade), teacher perceptions of their relationship with the child were assessed with the Student Teacher Relationship Scale (STRS; Pianta, 2001). The STRS is a widely used measure of teachers' perceptions of their relationships with students and has shown validity with regard to predicting academic and social functioning in pre-kindergarten through the elementary grades (Hamre & Pianta, 2001; Pianta et al., 2002). Scores can range from 1 (definitely does not apply) to 5 (definitely applies). This teacher self-report scale yields Closeness and Conflict relationship quality scores.

The Closeness scale is the mean of 7 items:

- I share an affectionate, warm relationship with this child.
- If upset, this child will seek comfort from me.
- This child values his/her relationship with me.
- When I praise this child, he/she beams with pride.
- This child spontaneously shares information about him/herself.
- It is easy to be in tune with what this student is feeling.
- The child openly shares his/her feelings and experience with me.

The Conflict scale is the mean of 8 items:

- This child and I always seem to be struggling with each other.
- This child is uncomfortable with physical affection or touch from me.
- This child easily becomes angry at me.
- This child remains angry or is resistant after being disciplined.
- Dealing with this child drains my energy.
- When this child arrives in a bad mood, I know we're in for a long and difficult day.
- This child's feelings toward me can be unpredictable or can change suddenly.
- This child is sneaky or manipulative with me.

Finally, we summarized the child's experiences regarding classroom practices using the Snapshot (see above for details) and family characteristics based on a short parent survey.

Summary of analysis variablesCHILD OUTCOMES (Pre-K)

Standardized Tests	Fall Pre-K				Spring Pre-K				Corr. Fall to Spring	Variable Name
	N	Mean	SD	α	N	Mean	SD	α	r	
PPVT	805	92.3	14.00	0.82	855	94.90	13.6	0.83	0.75	ppvtepf, ppvteps
WJ app problems SS	790	97.10	13.2	0.76	850	97.90	12.2	0.79	0.69	wj10_ssepf, wj10_sseps
WJ rhyming raw	798	1.6	2.5	0.73	852	2.80	3.50	0.64	0.65	wj21aepf, wj21aeps
2 or more		49%				68%				
OWLS	789	91.1	12.2	0.76	850	93.80	12.6	0.79	0.74	owlsepf, owlseps
Other Child Assessments	N	Mean	SD	α	N	Mean	SD	α	r	
Number of letters	804	7.1	8.5	0.97	853	12.0	9.4	0.97	0.80	lettere pf, lettere ps
2 or more %		62%				83%				
Number of numbers	804	4	3.8	0.95	853	6.20	3.7	0.93	0.75	numbere pf, numbere ps
2 or more %		59%				83%				
Teacher Ratings	N	Mean	SD	α	N	Mean	SD	α	r	
Hightower competence	927	3.41	0.75	0.95	949	3.58	0.77	0.95	0.66	htcomppf, htcompps
Hightower beh problems	927	1.53	0.54	0.92	946	1.48	0.54	0.92	0.62	htprobpf, htprobps
Language/Literacy	916	2.09	0.81	0.89	941	2.79	0.92	0.92	0.55	cslangpf, cslangps
Math reasoning					940	2.76	1.04	0.94		csmathps
STRS: Closeness					950	4.37	0.62	0.85		stcloseps
STRS: Conflict					950	1.62	0.72	0.87		stconflps
Spanish Assessment	N	Mean	SD	α	N	Mean	SD	α	r	
Naming letters	113	1.51	3.78		94	3.00	5.34		0.60	letterspf, lettersps
Naming numbers	113	1.46	2.75		94	3.45	3.43		0.55	numberspf, numbersps
TVIP stand score	112	78.46	10.12		92	82.86	14.8		0.53	tvips pf, tvipsps
Woodcock Muñoz	113	5.77	3.35		94	7.66	3.39		0.56	wm25sspf, wm25ssps
Family	N	Mean	SD	α	N	Mean	SD	α	r	
Poverty	932	0.56	0.50							poor p
Education	956	12.55	2.2							chmatednp
Household Size	975	4.3	1.34							chhsumnp

SNAPSHOT (Child level Pre-K)

	Fall Pre-K			Spring Pre-K			Corr. Fall to Spring	Variable Name
	N	Mean	SD	N	Mean	SD	r	
Activity Setting	900			946				
Free choice/center		0.35	0.16		0.33	0.15	0.61	asfcc_c_mpf, asfcc_c_mps
Routine		0.20	0.09		0.21	0.08	0.36	asrtn_c_mpf, asrtn_c_mps
Snacks/meals		0.13	0.08		0.14	0.07	0.50	asmls_c_mpf, asmls_c_mps
Small group		0.07	0.09		0.06	0.08	0.41	assgt_c_mpf, assgt_c_mps
Whole group		0.24	0.13		0.23	0.11	0.52	aswgt_c_mpf, aswgt_c_mps
Individual		0.02	0.04		0.04	0.05	0.28	asint_c_mpf, asint_c_mps
Adult Interactions	900			946				
Elaborated		0.10	0.08		0.08	0.06	0.27	aielb_c_mpf, aielb_c_mps
Routine/simple		0.12	0.09					airsm_mpf
Routine					0.01	0.02	0.03	airtn_c_mps
Simple					0.18	0.08	0.34	aismp_c_mps
None		0.69	0.13		0.72	0.11	0.42	ainone_c_mpf, ainone_c_mps
Child Engagement	900			946				
Read to		0.04	0.04		0.04	0.04	0.22	cerdg_mpf, cerdg_mps
Letter/Sound		0.03	0.05		0.03	0.04	0.31	celts_mpf, celts_mps
Oral Language		0.06	0.06		0.05	0.05	0.14	ceold_mpf, ceold_mps
Math		0.07	0.06		0.07	0.05	0.20	cemth_mpf, cemth_mps
Science		0.10	0.08		0.1	0.08	0.25	cesci_mpf, cesci_mps
Writing		0.01	0.02		0.01	0.02	0.13	cewrt_mpf, cewrt_mps
Social Studies		0.19	0.11		0.18	0.10	0.21	cescs_mpf, cescs_mps
Child Engagement (Mutually exclusive)	900			946				
Read to		0.06	0.05		0.05	0.04	0.15	ceread_c_mpf, ceread_c_mps
Letter/Sound		0.03	0.05		0.03	0.04	0.30	celts_c_mpf, celts_c_mps
Oral Language		0.06	0.06		0.05	0.05	0.12	ceold_c_mpf, ceold_c_mps
Math		0.05	0.05		0.06	0.05	0.22	cemth_c_mpf, cemth_c_mps
Science		0.07	0.07		0.07	0.06	0.24	cesci_c_mpf, cesci_c_mps
Writing		0.01	0.01		0.01	0.02	0.08	cewrt_c_mpf, cewrt_c_mps
Social Studies		0.13	0.09		0.12	0.08	0.17	cescs_c_mpf, cescs_c_mps

Teacher-Child Engagement	900			946				
Encourage/scaffolds		0.20	0.10					tcscf_mpf
Encourage					0.05	0.05		tcenc_mps
Scaffolds					0.05	0.05		tcscf_mps
Didactic		0.16	0.10		0.23	0.11	0.40	tcddt_mpf, tcddt_mps
Second language		0.02	0.05		0.02	0.05	0.75	tcscf_mpf, tcscf_mps

CHILD OUTCOMES (K)

Standardized Tests	Fall Pre-K				Spring Pre-K				Corr. Fall to Spring	Variable Name
	N	Mean	SD	α	N	Mean	SD	α	r	
PPVT	832	96.21	13.06	0.82	858	97.15	12.72	0.84	0.81	ppvtekf, ppvteks
WJ app problems SS	831	98.07	11.47	0.76	859	101.31	11.06	0.76	0.70	wj10_ssekf, wj10_sseks
WJ rhyming raw	830	4.79	4.22	0.73	858	7.18	4.50	0.68	0.61	wj21aekf, wj21aeks
2 or more %		79%				89%				
OWLS	830	94.06	12.60	0.82	857	94.69	12.93	0.73	0.79	owlsekf, owlseks
Other Child Assessments	N	Mean	SD	α	N	Mean	SD	α	r	
Number of letters	832	16.90	8.87	0.96	859	23.99	4.25	0.93	0.55	letterekf, lettereks
2 or more %		93%				100%				
Number of numbers	830	8.01	2.81	0.88	859	9.62	1.25	0.85	0.53	numberekf, numbereks
2 or more %		96%				100%				
Teacher Ratings	N	Mean	SD	α	N	Mean	SD	α	r	
Hightower competence	823	3.38	0.76	0.94	821	3.47	0.76	0.95	0.72	htcompkf, htcompks
Hightower beh problems	843	1.65	0.64	0.92	823	1.64	0.65	0.92	0.72	htprobkf, htprobks
Language/Literacy	827	2.40	0.92	0.92	817	3.66	0.95	0.94	0.60	cslangkf, cslangks
Math reasoning	822	2.39	0.96	0.94	818	3.68	0.92	0.94	0.45	cslmathkf, cslmathks
STRS: Closeness					822	4.25	0.66	0.85		stclosekf, stcloseks
STRS: Conflict					822	1.62	0.73	0.87		stconfkf, stconfks
Spanish Assessment	N	Mean	SD	α	N	Mean	SD	α	r	
Naming letters	64	8.14	8.35		29	17.07	9.36		0.50	letterskf, lettersks
Naming numbers	64	5.05	3.71		29	8.55	2.87		0.44	numberskf, numbersks
TVIP stand score	63	81.43	14.91		29	86.54	13.93		0.62	tvipskf, tvipsks
Woodcock Muñoz	64	72.72	14.99		29	74.07	16.61		0.73	wm25sskf, wm25ssks

Snapshot - Child Level (K)

	K			Variable Name
	N	Mean	SD	
Activity Setting	885			
Free choice/center		0.06	0.07	asfcc_mk, asfcc_mk
Routine		0.24	0.07	asrtn_mk, asrtn_mk
Snacks/meals		0.03	0.03	asmls_mk, asmls_mk
Small group		0.10	0.09	assgt_mk, assgt_mk
Whole group		0.37	0.10	aswgt_mk, aswgt_mk
Other		0.20	0.09	asoth_mk, asoth_mk
Adult Interactions	885			
Elaborated		0.11	0.06	aielb_mk, aielb_mk
Routine		0.01	0.01	airtn_mk, airtn_mk
Simple		0.18	0.05	aismp_mk, aismp_mk
None		0.70	0.09	ainone_mk, ainone_mk
Child Engagement	885			
Read to		0.07	0.04	ceread_mk, ceread_mk
Letter/Sound		0.10	0.05	celts_mk, celts_mk
Oral Language		0.06	0.04	ceold_mk, ceold_mk
Math		0.13	0.05	cemth_mk, cemth_mk
Science		0.07	0.05	cesci_mk, cesci_mk
Writing		0.06	0.40	cewrt_mk, cewrt_mk
Social Studies		0.12	0.08	cescs_mk, cescs_mk
Teacher - Child Engagement	885			
Encourage		0.19	0.08	tcenc_mk
Scaffolds		0.09	0.05	tcscf_mk
Didactic		0.43	0.10	tcddt_mk
Second language		0.02	0.07	tcslc_mk

Child Outcomes (Grade 1)

Teacher ratings	Fall K				Variable Name
	N	Mean	SD	α	
Hightower competence	595	3.59	0.85	0.96	htcompe
Hightower beh problems	595	1.66	0.71	0.93	htprobe
Language/Literacy	569	3.43	1.08	0.96	cslange
Math reasoning	557	3.53	1.03	0.95	csmathe
STRS: Closeness	594	4.28	0.70	0.87	stclosee
STRS: Conflict	594	1.62	0.77	0.88	stconfle

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